

StyleNova – Fashion Products E-Commerce Order Management System

1. Database Design

1.1 Review Table

Attribute	Data Type	Constraint
Review_ID	INT	Primary Key
Customer_ID	INT	Foreign Key
Product_ID	INT	Foreign Key
Review_Text	VARCHAR2(500)	NOT NULL
Review_Date	DATE	NOT NULL

1.2 Rating Table

Attribute	Data Type	Constraint
Rating_ID	INT	Primary Key
Customer_ID	INT	Foreign Key
Product_ID	INT	Foreign Key
Rating	INT	NOT NULL
Rating_Date	DATE	NOT NULL

3. Table Creation

```
SQL> CREATE TABLE StyleNova_Review (  
2     Review_ID NUMBER(10) PRIMARY KEY,  
3     Customer_ID NUMBER(10) NOT NULL,  
4     Product_ID NUMBER(10) NOT NULL,  
5     Review_Text VARCHAR2(500) NOT NULL,  
6     Review_Date DATE NOT NULL,  
7     FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID),  
8     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
9 );
```

Table created.

3.2 Create Rating Table

```
SQL> CREATE TABLE StyleNova_Rating (  
2     Rating_ID NUMBER(10) PRIMARY KEY,  
3     Customer_ID NUMBER(10) NOT NULL,  
4     Product_ID NUMBER(10) NOT NULL,  
5     Rating NUMBER(1) NOT NULL,  
6     Rating_Date DATE NOT NULL,  
7     FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID),  
8     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID),  
9     CHECK (Rating BETWEEN 1 AND 5)  
10 );
```

Table created.

SQL>

4. Review Data Insertion

```
SQL> INSERT INTO StyleNova_Review
  2 (Review_ID, Customer_ID, Product_ID, Review_Text, Review_Date)
  3 VALUES
  4 (1, 101, 101, 'Good quality and comfortable dress.', DATE '2026-09-20');

1 row created.

SQL>
SQL> INSERT INTO StyleNova_Review
  2 (Review_ID, Customer_ID, Product_ID, Review_Text, Review_Date)
  3 VALUES
  4 (2, 102, 102, 'Beautiful design and good fitting.', DATE '2026-09-21');

1 row created.

SQL>
SQL> INSERT INTO StyleNova_Review
  2 (Review_ID, Customer_ID, Product_ID, Review_Text, Review_Date)
  3 VALUES
  4 (3, 103, 103, 'The material is soft and comfortable.', DATE '2026-09-22');

1 row created.

SQL>
SQL> INSERT INTO StyleNova_Review
  2 (Review_ID, Customer_ID, Product_ID, Review_Text, Review_Date)
  3 VALUES
  4 (4, 104, 104, 'Nice colour and stylish design.', DATE '2026-09-23');

1 row created.
```

5. Rating Data Insertion

```
SQL> INSERT INTO StyleNova_Rating
  2 (Rating_ID, Customer_ID, Product_ID, Rating, Rating_Date)
  3 VALUES
  4 (2, 102, 102, 4, DATE '2026-09-21');

1 row created.

SQL>
SQL> INSERT INTO StyleNova_Rating
  2 (Rating_ID, Customer_ID, Product_ID, Rating, Rating_Date)
  3 VALUES
  4 (3, 103, 103, 5, DATE '2026-09-22');

1 row created.

SQL>
SQL> INSERT INTO StyleNova_Rating
  2 (Rating_ID, Customer_ID, Product_ID, Rating, Rating_Date)
  3 VALUES
  4 (4, 104, 104, 4, DATE '2026-09-23');

1 row created.

SQL>
SQL> INSERT INTO StyleNova_Rating
  2 (Rating_ID, Customer_ID, Product_ID, Rating, Rating_Date)
  3 VALUES
  4 (5, 105, 105, 5, DATE '2026-09-24');
INSERT INTO StyleNova_Rating
```

6. Retrieve Product Review Details

```
SQL> SELECT
  2     R.Review_ID,
  3     R.Customer_ID,
  4     R.Product_ID,
  5     P.Product_Name,
  6     R.Review_Text,
  7     R.Review_Date
  8 FROM StyleNova_Review R
  9 JOIN Product P
 10     ON R.Product_ID = P.Product_ID
 11 ORDER BY R.Review_Date;
```

REVIEW_ID	CUSTOMER_ID	PRODUCT_ID
-----------	-------------	------------

PRODUCT_NAME

REVIEW_TEXT

REVIEW_DA

1	101	101
Cotton T-Shirt		
Good quality and comfortable dress.		
20-SEP-26		

REVIEW_ID	CUSTOMER_ID	PRODUCT_ID
-----------	-------------	------------

PRODUCT_NAME

REVIEW_TEXT

REVIEW_DA

2	102	102
Denim Jeans		
Beautiful design and good fitting.		
21-SEP-26		

REVIEW_ID	CUSTOMER_ID	PRODUCT_ID
-----------	-------------	------------

PRODUCT_NAME

REVIEW_TEXT

REVIEW_DA

3	103	103
Floral Kurti		
The material is soft and comfortable.		
22-SEP-26		

REVIEW_ID	CUSTOMER_ID	PRODUCT_ID
-----------	-------------	------------

PRODUCT_NAME

REVIEW_TEXT

REVIEW_DA

4	104	104
Kids Hoodie		
Nice colour and stylish design.		
23-SEP-26		

7. Calculate Average Product Ratings

```
SQL> SELECT
  2     Product_ID,
  3     AVG(Rating) AS Average_Rating
  4 FROM StyleNova_Rating
  5 GROUP BY Product_ID
  6 ORDER BY Average_Rating DESC;
```

PRODUCT_ID	AVERAGE_RATING
------------	----------------

103	5
101	5
104	4
102	4

8. Identify Highly Rated Products

```
SQL> SELECT
  2     P.Product_ID,
  3     P.Product_Name,
  4     AVG(R.Rating) AS Average_Rating
  5 FROM Product P
  6 JOIN StyleNova_Rating R
  7   ON P.Product_ID = R.Product_ID
  8 GROUP BY P.Product_ID, P.Product_Name
  9 ORDER BY Average_Rating DESC;

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      103
Floral Kurti      5

      101
Cotton T-Shirt    5

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      104
Kids Hoodie       4

      102
Denim Jeans       4

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      4
```

9. Product Rating Report

```
SQL> SELECT
  2     P.Product_ID,
  3     P.Product_Name,
  4     AVG(R.Rating) AS Average_Rating
  5 FROM Product P
  6 JOIN StyleNova_Rating R
  7   ON P.Product_ID = R.Product_ID
  8 GROUP BY P.Product_ID, P.Product_Name
  9 ORDER BY Average_Rating DESC;

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      103
Floral Kurti      5

      101
Cotton T-Shirt    5

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      104
Kids Hoodie       4

      102
Denim Jeans       4

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      4
```